GE VERNOVA

GridOS Simulation Framework

0.21.0

Installation Guide

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Change History

Date	Change Description
2024-Sep-17	GridOS Simulation Framework 0.1.0 (24R09): • This is a new document for the initial internal release of GridOS Simulation Framework.
2024-Sep-26	GridOS Simulation Framework 0.2.0 (24R10): • Added GridOS Data Fabric Prerequisites • Simulation Manager replaces Organizer with Simulation Plugin • Added ArgoCD install steps • Added GridOS Simulation Framework Uninstall
2024-Nov-11	GridOS Simulation Framework 0.3.0 (24R11): • Updated GridOS Data Fabric Prerequisites • Require Helm 3.16.2 or later
2024-Nov-15	GridOS Simulation Framework 0.4.0 (24R12): • Add section on Kafka related helm values for Simulation Framework helm charts • Clarify use of `--values-literal-file` with ArgoCD
2024-Dec-13	GridOS Simulation Framework 0.5.0 (24R13): • Use markets-kafka-helm to install Kafka

2025-Jan-24	GridOS Simulation Framework 0.6.0 (25R01):
	• `simulation` namespace replaced with `foundation-env-default`
	• `simulation-worker` namespace replaced with `simulation-grid-operate`, `simulation-grid-plan` and `simulation-grid-transact`
	• Helm umbrella charts renamed:
	  ○ `simulation-charts` renamed to `simulation-apps`
	  ○ `simulation-worker-charts` renamed to `simulation-support`
	• Strict mTLS enabled by default

2025-Feb-27	GridOS Simulation Framework 0.7.0 (25R02): • Install Timebase with Data Fabric
2025-Mar-24	GridOS Simulation Framework 0.8.0 (25R03): • Remove support for UAA

Date	Change Description
2025-Apr-28	GridOS Simulation Framework 0.9.0 (25R04): • Simulation Manager migrated from Organizer Helm chart to standalone Helm chart. • Simulation Manager REST API path changed to `/simulation/manager/v1` • Add Authorization to Simulation Manager API
2025-Aug-29	GridOS Simulation Framework 0.14.0 (25R09): • Remove hello-d from Data Fabric prerequisites
2025-Oct-03	GridOS Simulation Framework 0.15.0 (25R10): • Update Data Fabric overlays
2025-Nov-10	GridOS Simulation Framework 0.16.0 (25R11): • Update Data Fabric overlays • Update Kafka documentation to use KRaft and markets-kafka default values • Separate Archive Manager dependency for installation
2025-Nov-18	GridOS Simulation Framework 0.17.0 (25R12): • Update Data Fabric overlay to workaround Timebase REST out of memory error
2025-Dec-16	GridOS Simulation Framework 0.18.0 (25R13): • Remove workaround for Timebase REST out of memory error from Data Fabric overlay
2026-Jan-13	GridOS Simulation Framework 0.19.0 (26R01): • Update minimum required versions of helm and kubectl
2026-Feb-27	GridOS Simulation Framework 0.20.0 (26R02): • Update market-kafka version to 2.3.0
2026-Mar-19	GridOS Simulation Framework 0.21.0 (26R03): • Replace `dig-grid-artifactory.apps.ge.com` with `gart.software.gevernova.com` • Use `virtual-helm` repository to install all Helm charts • Tested with ArgoCD 3.2.1

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1. Introduction

GridOS Simulation Framework provides the core services and Kubernetes resources for GridOS products to take part in coordinated simulation. It runs on top of GridOS Foundation and utilizes components of GridOS Data Fabric.

1.1. Preparing for Installation

Currently, this guide is intended for internal use only. The following assumptions are made:

- A GridOS Foundation environment is available.
  - See GridOS Foundation Prerequisites for details on the required configuration for this environment.
- GridOS Data Fabric is installed
  - See GridOS Data Fabric Prerequisites for details on the required configuration.
- Access to the following Docker registries on Grid Artifactory:
  - `gart.software.gevernova.com/foundation-docker`
  - `gart.software.gevernova.com/gridos-shared-services-docker-virtual`
- Access to the following Helm chart repositories on Grid Artifactory:
  - `https://gart.software.gevernova.com/artifactory/virtual-helm`
- Helm (v3.17.0 or later) is available with access to your GridOS Foundation environment
- kubectl (v1.35.2 or later) is available with access to your GridOS Foundation environment
- Optionally, ArgoCD CLI (v3.2.1) if using ArgoCD

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2. GridOS Foundation Prerequisites

GridOS Simulation Framework has been tested on a single node GridOS Foundation environment. However, there should be no limitations running in a multi-node environment.

This release is certified against the version of GridOS Foundation listed in Software Compatibility.

> When GridOS Foundation Multi-Environment support is available it is intended that GridOS Simulation will become a dedicated environment.

2.1. Overlay Values required by GridOS Simulation Framework

Install GridOS Foundation according to its documentation. Only GridOS Foundation Base is required.

Assuming the standard Helm profiles are used from foundation-reference-app, GridOS Simulation Framework requires that the following additional overlay values be used with GridOS Foundation Base.

```yaml
global:
  minio:
    enabled: true

strimzi-kafka-operator:
  enabled: true

minio-operator:
  enabled: true

# For dev environments
openldap:
  enabled: true
```

Note that GridOS Simulation Framework tests are run on an environment where `global.mtls.enabled = true`, but this setting can be configured as desired.

2.2. Roles and Permissions required by GridOS Simulation Framework

GridOS Simulation Framework defines a number of permissions to support authorization.

GridOS Foundation roles and permissions **must** be configured to ensure that a user groups are configured to include the required GridOS Simulation Framework permissions.

See example Foundation Data entries for defining GridOS Simulation Framework permissions.

2.3. Add virtual-helm Repository

Kafka and Archive Manager charts are available in the `virtual-helm` repository on Grid Artifactory at `https://gart.software.gevernova.com/artifactory/api/helm/virtual-helm`. This repository should be added to your Helm configuration to allow installation of these components. `${GRIDART_USR}` and `${GRIDART_PSW}` environment variables are set to a Grid Artifactory username and password that have access to the `virtual-helm` repository.

```
helm repo add virtual-helm https://gart.software.gevernova.com/artifactory/api/helm/virtual-helm
--username ${GRIDART_USR} --password ${GRIDART_PSW}
```

2.4. Kafka

Kafka with mTLS and simple ACLs enabled (as required by ADR-23 Kafka Zerotrust) is required for GridOS Simulation Framework.

If you already have a Kafka broker installed for other purposes, it can be used. Otherwise, Kafka can be deployed via Helm using markets-kafka-helm.

Configure Kafka as required for your environment. The Helm value overrides for a simple single node Kafka broker are as follows

```yaml
markets-kafka:
  kraft:
    enabled: true

# Single combined node (controller + broker):
  kafkaNodePools:
    - name: combined
      roles:
        - controller
        - broker
      replicas: 1
      storage:
        type: jbod
        volumes:
          - id: 0
            type: persistent-claim
            size: 100Gi
            kraftMetadata: shared
            deleteClaim: false

  kafka:
    authorization:
      enabled: true
    version: 3.9.0
```

```
    # Override replication factors for single-node setup
    config:
      default.replication.factor: 1
      offsets.topic.replication.factor: 1
      transaction.state.log.replication.factor: 1
      transaction.state.log.min.isr: 1
      min.insync.replicas: 1
    resources:
      limits:
        cpu: 2
        memory: 2G
```

By default, GridOS Foundation watches the `foundation-env-default` namespace for Kafka resources and the Helm chart can be installed as follows.

```
# Version 2.3.0 is currently used in Simulation Framework testing.
helm -n foundation-env-default upgrade -i kafka virtual-helm/markets-kafka --version 2.3.0 -f kafka-values.yaml
```

2.5. Archive Manager

GridOS Simulation Framework requires Archive Manager to be installed and available. Archive Manager can be installed via Helm as follows.

To match the SIRE environment it is assumed that Archive Manager is installed in the `am` namespace.

```
# Version 0.17.2 is currently used in Simulation Framework testing.
helm -n am upgrade -i archive-manager virtual-helm/archive-manager --version 0.17.2 -f am-values.yaml
```

An example `am-values.yaml` file is as follows

```
global:
  clusterExternalUrl: https://ip-10-227-48-188.ec2.internal
  zitadel:
    enabled: true

archive:
  createbucket: true
  bucketName: simulation-bucket
image:
  registryUrl: gart.software.gevernova.com
  pullPolicy: IfNotPresent
```

3. GridOS Data Fabric Prerequisites

This release is certified against the version of GridOS Data Fabric listed in Software Compatibility.

4. Overlay Values required by GridOS Simulation Framework

Install GridOS Data Fabric according to its documentation.

GridOS Simulation Framework requires the following GridOS Data Fabric services

- Timebase (With the REST API enabled)

GridOS Simulation Framework tests are run against a minimal GridOS Data Fabric installed with the following overlays

```yaml
global:
  mtls:
    enabled: true
  zitadel:
    enabled: true

anybase:
  enabled: false

consul:
  enabled: false

organizer:
  enabled: false

superset:
  enabled: false

discover:
  enabled: false

hello-d:
  enabled: false

# REST API is enabled by default
timebase:
  enabled: true
  # Disable metadata as Data Hub is disabled
  metadata:
    enabled: false
  flight:
    enabled: false
  rest:
    # Set limits as REST container so that it will have max heap size.
    resources:
      requests:
        memory: 1Gi
      limits:
        memory: 1Gi
```


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```yaml
mapservice:
  enabled: false

cns:
  enabled: false

explore:
  enabled: false
```

`global.mtls.enabled` can be configured as desired.

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5. GridOS Simulation Framework Installation

GridOS Simulation Framework can be installed using either Helm or ArgoCD to suit your overall deployment pipeline. In either case the simulation namespaces must first be created.

> When GridOS Foundation Multi-Environment support is available it is intended that GridOS Simulation will become a dedicated environment and use namespace `foundation-env-simulation` instead of `foundation-env-default`.

5.1. Create the simulation namespaces

A number of namespaces are created to support simulation.

- `simulation-grid-operate`
- `simulation-grid-plan`
- `simulation-grid-transact`

These are the namespaces that GridOS products install their services in a simulation environment. They are also where `ON_DEMAND` services for ephemeral simulations are deployed.

Enable Istio on all namespaces. Create a file with the content below:

```yaml
apiVersion: v1
kind: Namespace
metadata:
  name: simulation-grid-operate
  labels:
    istio-injection: enabled
---
apiVersion: v1
kind: Namespace
metadata:
  name: simulation-grid-plan
  labels:
    istio-injection: enabled
---
apiVersion: v1
kind: Namespace
metadata:
  name: simulation-grid-transact
  labels:
    istio-injection: enabled
```

and apply using `kubectl apply -f simulation-namespaces.yaml`

Kyverno policies will automatically mutate the default service accounts to set `automountServiceAccountToken: false`

5.2. Helm

5.2.1. Add Helm repository

The GridOS Shared Services Helm repository from Grid Artifactory must be added. `${GRIDART_USR}` and `${GRIDART_PSW}` environment variables are set to a Grid Artifactory username and password that have access to this repository.

```
helm repo add virtual-helm https://gart.software.gevernova.com/artifactory/api/helm/virtual-helm
--username ${GRIDART_USR} --password ${GRIDART_PSW}
```

5.2.2. Simulation Global Values

See Simulation Helm Global Values for details of the standard global values used across GridOS Simulation Framework Helm charts.

If GridOS Data Fabric has not been installed in the default `data-fabric` namespace then you will need to override the global value for the Data Fabric namespace.

5.2.3. Kafka Helm Values

Simulation Framework charts follow the GridOS Helm Global Conventions to connect to Kafka. The default values are shown below

```
global:
  kafka:
    strimzi:
      localClusterResourceName: markets-kafka
      createUsers: true
    localCluster:
      port: 9093
      namespace: "{{ .Values.global.foundation.envNamespace }}"
      authorizationType: simple
```

You will need to override as appropriate to connect to the Kafka broker running in your environment.

5.2.4. Archive Manager

Simulation Framework assumes that Archive Manager is installed and available via the following values.

```
global:
  simulation:
```

```
    archiveManager:
      address: "archive-manager.am"
      port: 8080
```

You will need to override these values if Archive Manager is installed in a different namespace or using different service name or port.

5.2.5. Install Simulation Framework Charts

GridOS Simulation Framework comprises three Helm charts

- simulation-crds
  - GridOS Simulation CRDs
- simulation-apps
  - Simulation Manager
  - Simulation Time Service
- simulation-support
  - Kubernetes resources to support `ON_DEMAND` deployment

Review the default values for simulation-apps and simulation-support to check if you need to override any other values. Note that

- `global.clusterExternalUrl` **must** be set to the value for your environment.
- The default setting for `global.mtls.enabled` is `true`, but this setting can be configured as desired for your environment.

The Helm commands below are used to install all Simulation Framework related charts. See component bill of materials for the versions to use for `${CRD_VERSION}`, `${SIMULATION_APPS_VERSION}` and `${SIMULATION_SUPPORT_VERSION}`. Add additional value override arguments as required.

> If upgrading from 0.9.0 (25R05) or older you first need to uninstall the complete Simulation Framework due to changes to Kubernetes resources labels done according to "146-ADR Standard labels on Kubernetes Resources via Helm charts". See GridOS Simulation Framework Uninstall In addition to the uninstall steps, delete the simulation-tests using
>
> ```
> helm -n foundation-env-default delete simulation-tests
> ```

```
# Install the simulation CRDs chart to the foundation-env-default namespace
helm upgrade -n foundation-env-default -i simulation-crds virtual-helm/simulation-crds --version
${CRD_VERSION}
# Install the simulation-apps umbrella chart to the foundation-env-default namespace
helm upgrade -n foundation-env-default -i simulation-apps virtual-helm/simulation-apps --version
${SIMULATION_APPS_VERSION} --set global.clusterExternalUrl=https://ip-10-227-48-188.ec2.internal
# Install the simulation-support umbrella chart to the application and simulator namespaces
```

```
helm upgrade -n simulation-grid-operate -i simulation-support virtual-helm/simulation-support
--version ${SIMULATION_SUPPORT_VERSION} --set global.clusterExternalUrl=https://ip-10-227-48-
188.ec2.internal
helm upgrade -n simulation-grid-plan -i simulation-support virtual-helm/simulation-support --version
${SIMULATION_SUPPORT_VERSION} --set global.clusterExternalUrl=https://ip-10-227-48-188.ec2.internal
helm upgrade -n simulation-grid-transact -i simulation-support virtual-helm/simulation-support
--version ${SIMULATION_SUPPORT_VERSION} --set global.clusterExternalUrl=https://ip-10-227-48-
188.ec2.internal
```

5.3. ArgoCD

ArgoCD applications can be created by using the GridOS Simulation Framework Helm charts directly.

The commands below assume

- Simulation namespaces have been created
- An ArgoCD instance is available to use with your GridOS environment
- The ArgoCD CLI is installed and a user is logged in.

5.3.1. Add Helm Repository

The GridOS Shared Services Helm repository from Grid Artifactory must be added. `${GRIDART_USR}` and the `${GRIDART_PSW}` environment variables are set to a Grid Artifactory username and password that have access to this repository.

```
argocd repo add https://gart.software.gevernova.com/artifactory/api/helm/virtual-helm --type helm
--name virtual-helm --username ${GRIDART_USR} --password ${GRIDART_PSW}
```

5.3.2. Install Simulation Applications

ArgoCD applications can be created by using the GridOS Simulation Framework Helm charts directly.

See component bill of materials for the versions to use for `${CRD_VERSION}`, `${SIMULATION_APPS_VERSION}` and `${SIMULATION_SUPPORT_VERSION}`.

You will need to create `sim-apps-values.yaml` and `sim-support-values.yaml` files to pass as the argument for `--values-literal-file` with the relevant Helm chart. Refer to Helm for details on what values can or should be overridden. For example, you will need additional values if using Zitadel/Apisix and/or a Kafka broker that differs from the default.

As a minimum the files will include

```
global:
```

```
  clusterExternalUrl: https://ip-10-227-48-188.ec2.internal ①
```

① Value should reference your hostname

The ArgoCD commands below can be used to install Simulation Framework.

```
# simulation-crds
argocd app create simulation-crds --repo
https://gart.software.gevernova.com/artifactory/api/helm/virtual-helm --helm-chart simulation-crds
--revision ${CRD_VERSION} --dest-namespace foundation-env-default --dest-server
https://kubernetes.default.svc --sync-policy auto --self-heal --allow-empty --auto-prune
# simulation-apps
argocd app create simulation-apps --repo
https://gart.software.gevernova.com/artifactory/api/helm/virtual-helm --helm-chart simulation-apps
--revision ${SIMULATION_APPS_VERSION} --dest-namespace foundation-env-default --dest-server
https://kubernetes.default.svc --values-literal-file sim-apps-values.yaml --sync-policy auto --self
-heal --allow-empty --auto-prune
# simulation-support
argocd app create simulation-grid-operate-support --repo
https://gart.software.gevernova.com/artifactory/api/helm/virtual-helm --helm-chart simulation-support
--revision ${SIMULATION_SUPPORT_VERSION} --dest-namespace simulation-grid-operate --dest-server
https://kubernetes.default.svc --values-literal-file sim-support-values.yaml --sync-policy auto --self
-heal --allow-empty --auto-prune
argocd app create simulation-grid-plan-support --repo
https://gart.software.gevernova.com/artifactory/api/helm/virtual-helm --helm-chart simulation-support
--revision ${SIMULATION_SUPPORT_VERSION} --dest-namespace simulation-grid-plan --dest-server
https://kubernetes.default.svc --values-literal-file sim-support-values.yaml --sync-policy auto --self
-heal --allow-empty --auto-prune
argocd app create simulation-grid-transact-support --repo
https://gart.software.gevernova.com/artifactory/api/helm/virtual-helm --helm-chart simulation-support
--revision ${SIMULATION_SUPPORT_VERSION} --dest-namespace simulation-grid-transact --dest-server
https://kubernetes.default.svc --values-literal-file sim-support-values.yaml --sync-policy auto --self
-heal --allow-empty --auto-prune
```

5.4. External HTTP Clients

External HTTP clients can access the Simulation Framework APIs via GridOS Foundation Ingress using OAuth2 Client Credentials grant type.

When using Zitadel a user is created via a `ConfigMap`. Create the file `simulation-user-zitadel.yaml` with the content below

```yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: simulation-user
  labels:
    zitadel.gevernova.com/track: "true"
data:
  name: "simulation-user"
  org: foundation
```

```
  secretName: "simulation-user-secret"
  groups: |-
    - sim.admin ①
```

① This group name must map to one of the user groups available on GridOS Foundation that includes the appropriate Simulation Framework permissions.

The user can be installed

```
kubectl -n foundation-env-default apply -f simulation-user-zitadel.yaml
```

A secret named `simulation-user-secret` will be created in the same namespace where the user was installed.

The client ID, client secret and project ID for the created user can be extracted using the following commands

```
CLIENT_ID=`kubectl -n foundation-env-default get secret simulation-user-secret -o yaml | grep clientId
| awk 'NR==1{print $2}' | base64 -d`
CLIENT_SECRET=`kubectl -n foundation-env-default get secret simulation-user-secret -o yaml | grep
clientSecret | awk 'NR==1{print $2}' | base64 -d`
PROJECT_ID=`kubectl -n foundation-cluster-zerotrust get configmap zitadel-projects -o jsonpath
='{.data.foundation\.foundation\.id}'`

echo Client ID $CLIENT_ID
echo Client Secret $CLIENT_SECRET
echo Project ID $PROJECT_ID
```

Alternatively see accessing secured API's from external systems in the GridOS Foundation documentation.

```
# Get the token
# Replace the project ID number, client ID and client secret value with those from your system
curl -k --request POST \
  --url https://zitadel.ip-10-227-48-188.ec2.internal/oauth/v2/token \
  --header 'Content-Type: application/x-www-form-urlencoded' \
  --data grant_type=client_credentials \
  --data scope='openid profile email urn:zitadel:iam:org:project:id:281588110947516510:aud
read:messages' \
  --user "simulation-
user@foundation.local:RzKjdxp2Q4abByqBEtE72KQEFcQM1cfTcoVrlgrREkzefVwSNsJmMabGqbfjAHRh"
# Set env var for subsequent calls
export TOKEN="Bearer eyJhbGci...."
# Example calling discover endpoint on Simulation Manager
curl -k -X GET -H "Authorization: $TOKEN" https://service.ip-10-227-48-
188.ec2.internal/simulation/manager/v1/discover
```

6. GridOS Simulation Framework Uninstall

GridOS Simulation Framework can be uninstalled from your GridOS environment using the commands shown below.

> Ensure that any Simulator or Application CRs are uninstalled **before** removing the Simulation CRDs themselves.

6.1. Helm

```
helm -n simulation-grid-operate delete simulation-support
helm -n simulation-grid-plan delete simulation-support
helm -n simulation-grid-transact delete simulation-support
helm -n foundation-env-default delete simulation-apps
helm -n foundation-env-default delete simulation-crds
```

6.2. ArgoCD

```
argocd app delete simulation-grid-operate-support --yes
argocd app delete simulation-grid-plan-support --yes
argocd app delete simulation-grid-transact-support --yes
argocd app delete simulation-apps --yes
argocd app delete simulation-crds --yes
# ArgoCD does not delete CRDs by design so you need to do this via kubectl
kubectl delete crd applications.simulation.gridos.com
kubectl delete crd simulators.simulation.gridos.com
```

6.3. Namespaces

```
kubectl delete -f simulation-namespaces.yaml
# or
kubectl delete namespace simulation-grid-operate
kubectl delete namespace simulation-grid-plan
kubectl delete namespace simulation-grid-transact
```

7. Example GridOS Foundation Data

Below are example snippets of roles and permissions that could be added to the files in a foundation-data Helm chart to support GridOS Simulation Framework.

7.1. roles-config-v2.yaml

```yaml
usergroups:

  # append after existing usergroups entries

  - groupName: Simulation Administrator
    description: Simulation Administrators
    mappedGroupName: sim.admin

  - groupName: Simulation Operator
    description: Simulation Operators
    mappedGroupName: sim.operator

  - groupName: Simulation Trainee
    description: Simulation Trainees
    mappedGroupName: sim.trainee

  - groupName: Simulation Instructor
    description: Simulation Instructors
    mappedGroupName: sim.instructor

roles:

  # append after existing roles entries

  - roleName: Simulation Scenario Read
    description: Read a users simulation scenarios

  - roleName: Simulation Scenario Write
    description: Write a users simulation scenarios

  - roleName: Simulation Scenario Read All
    description: Read all simulation scenarios

  - roleName: Simulation Scenario Write All
    description: Write all simulation scenarios

  - roleName: Simulation Instance Read
    description: Read a users simulation instances

  - roleName: Simulation Instance Read All
    description: Read all simulation instances

  - roleName: Simulation Instance Execute
```

```
    description: Execute a users simulation instances

  - roleName: Simulation Instance Execute All
    description: Execute all simulation instances

permissions:

  # append after existing permissions entries

  - permissionName: sim.scenario.read
    description: Read access to my scenarios
  - permissionName: sim.scenario.readAll
    description: Read access to all scenarios
  - permissionName: sim.scenario.write
    description: Write access to my scenarios
  - permissionName: sim.scenario.writeAll
    description: Write access to all scenarios
  - permissionName: sim.instance.read
    description: Read access to my instances
  - permissionName: sim.instance.readAll
    description: Read access to all instances
  - permissionName: sim.instance.execute
    description: Execute access to my instances
  - permissionName: sim.instance.executeAll
    description: Execute access to all instances

mapping:
  usergroups:

  # append after existing usergroups entries

  - name: Simulation Administrator
    roles: ['Simulation Scenario Read All', 'Simulation Scenario Write All', 'Simulation Instance Read
All', 'Simulation Instance Execute All']

  - name: Simulation Operator
    roles: ['Simulation Scenario Read', 'Simulation Scenario Write', 'Simulation Instance Read',
'Simulation Instance Execute']

  - name: Simulation Instructor
    roles: ['Simulation Scenario Read All', 'Simulation Scenario Write All', 'Simulation Instance
Read', 'Simulation Instance Execute']

  - name: Simulation Trainee
    roles: ['Simulation Scenario Read All', 'Simulation Instance Read', 'Simulation Instance Execute']

roles:

  # append after existing roles entries

  - name: Simulation Scenario Read
    permissions: [
      'sim.scenario.read'
```

```
    ]

  - name: Simulation Scenario Write
    permissions: [
      'sim.scenario.write'
    ]

  - name: Simulation Scenario Read All
    permissions: [
      'sim.scenario.readAll'
    ]

  - name: Simulation Scenario Write All
    permissions: [
      'sim.scenario.writeAll'
    ]

  - name: Simulation Instance Read
    permissions: [
      'sim.instance.read'
    ]

  - name: Simulation Instance Read All
    permissions: [
      'sim.instance.readAll'
    ]

  - name: Simulation Instance Execute
    permissions: [
      'sim.instance.execute'
    ]

  - name: Simulation Instance Execute All
    permissions: [
      'sim.instance.executeAll'
    ]
```

7.2. OpenLDAP

If you are using OpenLDAP on a development environment then you'll need to add the Simulation Framework groups to the custom LDIF ConfigMap.

Below is an example showing the groups detailed above.

```
dn: cn=sim.admin,ou=nstar_ou,{{ LDAP_BASE_DN }}
objectClass: posixGroup
objectClass: top
cn: sim.admin
gidNumber: 540
memberUid: cn=simadmin1,ou=users,{{ LDAP_BASE_DN }}
```

```
dn: cn=sim.operator,ou=nstar_ou,{{ LDAP_BASE_DN }}
objectClass: posixGroup
objectClass: top
cn: sim.operator
gidNumber: 550
memberUid: cn=simops1,ou=users,{{ LDAP_BASE_DN }}

dn: cn=sim.trainee,ou=nstar_ou,{{ LDAP_BASE_DN }}
objectClass: posixGroup
objectClass: top
cn: sim.trainee
gidNumber: 560
memberUid: cn=simtrainee1,ou=users,{{ LDAP_BASE_DN }}

dn: cn=sim.instructor,ou=nstar_ou,{{ LDAP_BASE_DN }}
objectClass: posixGroup
objectClass: top
cn: sim.instructor
gidNumber: 570
memberUid: cn=siminstructor1,ou=users,{{ LDAP_BASE_DN }}
```

In this case it is assumed new users would also be added to the custom LDIF ConfigMap but you could also map the new groups to existing users.